Plant-AT: Customized Hybrid Attention Model for Diagnosing Plant Disease

Akash Nagappagol and Suneetha Budihal

School of Electronics and Communication Engineering, KLE Technological University, Hubballi, Karnataka, INDIA https://www.kletech.ac.in/

Abstract. This paper presents Plant-AT, a novel hybrid attention model built for real-time plant disease categorization, which addresses a critical agricultural concern. Crop diseases in India cause yearly losses of 10-30% in tomatoes and 15-40% in grapes, putting food security at risk. Plant-AT combines Inverted Residual Blocks, an innovative LG-Attention Transformer, and a Robust Multi-Scale Fusion Mechanism, with key contributions such as Separable Self-Attention (SSA) for better global context understanding and Neighborhood Attention (NA) for precise local feature extraction. Plant-AT, with controlled 8.8 million trainable parameters, strikes a compromise between accuracy and efficiency, making it appropriate for deployment on edge devices with limited resources.This paper also introduces hybrid loss function which addresses class imbalance issues which are common in plant disease dataset.

Keywords: Neighborhood Attention(NA) · Separable Self-Attention(SSA) · Robust multi-Scale Fusion · Hybrid loss Function · Edge Deployment.

1 Introduction

Transformers have made a tremendous influence on AI research, beginning with their success in machine translation and Natural Language Processing, and later expanding to modalities like audio and vision. The universal architecture of transformers, predicated on attention techniques, has contributed to their widespread acceptance across numerous AI areas. In computer vision, the original transformer design [1] has been extended for image classification and dense tasks, gradually surpassing Convolutional Neural Networks (CNNs) as the actual standard.This innovation is especially important in agriculture, where vision applications improve crop management and disease surveillance.

In India, reliably classifying plant diseases is critical for increasing agricultural output and sustainability. For crops such as tomato and grape, early and precise disease identification can reduce production loss and the need for chemical treatments, which is both economically and environmentally advantageous. Vision Transformers [2], with their superior attention processes, provide substantial advantages in this situation. They excel at catching global interdependencies within images, which is critical for detecting subtle disease symptoms that can change across different sections of the plant.

Despite these advantages, scalability of self-attention mechanisms is still difficult due to their quadratic complexity in relation to input size and lack of local inductive bias. This can be especially challenging in vision applications that require high-resolution images of plant leaves. Nonetheless, continuous research and development in transformer architectures remove these constraints, making them more suited for large-scale agricultural applications. Researchers can use these advancements to create more effective disease categorization and crop management tools, resulting in better agricultural practices and food security. Towards this, Our main contributions are,

- We introduce a novel hybrid attention-based architecture for plant disease classification that effectively distinguishes between plant species and disease types. To improve the dataset, we applied data augmentation and addressed class imbalance using a WeightedRandomSampler.
- We proposed a hybrid loss function, to balance learning between easy and challenging plant classes and disease classes. For optimization, we utilized the AdamW optimizer with a cosine annealing scheduler to ensure robust and efficient convergence. We benchmarked our architecture against state-of-the-art image classification systems, demonstrating its superior performance.

Section 2 encompasses the related work that significantly contributed to the design of the Plant-AT Model. Section 3 elaborates on a novel hybrid attention architecture. Section 4 delineates the performance metrics and training processes employed. Section 5 discusses the dataset utilized for model training and validation. Finally, Sections 6, 7 and 8 present the results, conclusion and acknowledgement, respectively.

2 Related work

Research [3] [4] explores AI and image processing techniques for plant disease classification, focusing on CNNs and transfer learning. It evaluates datasets based on quality, diversity, and capture conditions, addressing challenges like symptom complexity, high-quality labeled data requirements, environmental constraints, and the need for lightweight models in real-time applications. Additionally, deep learning in precision agriculture, including crop management, disease diagnostics, and precision irrigation, is examined. Study [5] discusses Vision Transformers (ViT) and Hybrid Vision Transformers (HVT) in computer vision. ViTs utilize self-attention on image patches for global visual patterns, outperforming CNNs in object recognition, while HVTs merge CNN local feature extraction with ViT global context modeling to enhance image recognition tasks, addressing ViT limitations such as small datasets and processing inefficiencies. The study encourages further exploration of hybrid models for edge device applications. The Research paper [12] introduces a hybrid model that merges the CNN's local feature extraction with the ViT's global context capabilities. With only 0.85 million parameters, this lightweight model is well-suited for IoT-based agriculture and resource-limited settings. It achieves superior results on five datasets, outperforming nine state-of-the-art methods, while explainability tools like Grad-CAM and LIME validate its disease localization accuracy. However, the model's compact size may limit predictive robustness in complex, real-time environments requiring high feature detail. Additionally, its reliance on single-leaf images in controlled settings could hinder performance in dynamic, multi-leaf contexts typical of practical applications.

Research [6] compares ViTs and hybrid models, emphasizing efficiency and error rates. ViTs capture global dependencies via self-attention, outperforming CNNs in various tasks. Hybrid models improve performance by combining CNN and ViT capabilities. The study introduces the Efficient Error Rate (EER) metric, factoring in trainable parameters and FLOPs, and highlights techniques like Token Merging and dynamic inference algorithms (A-ViT, SPViT) to boost computing efficiency for real-time applications. Significant improvements in transformer efficiency and scalability are proposed in studies [7] [8], with the Neighborhood Attention Transformer (NAT) and its Neighborhood Attention (NA) mechanism reducing complexity from quadratic to linear by focusing self-attention on nearest neighbors, maintaining locality and translational equivariance. Efficient NA implementations using C++ and CUDA achieve up to 40% faster performance with less memory usage compared to traditional methods like Swin Transformers [9]. NAT's hierarchical architecture delivers competitive image classification results.
A novel Separable Self-Attention (SSA) mechanism is introduced in [10] to enhance efficiency on resource-constrained devices. SSA reduces the quadratic complexity of traditional Multi-Headed Attention (MHA) to linear by computing context scores, relative to a latent token and significantly lowers processing and memory overhead, while maintaining competitive performance and improved inference speed.The study [11] provides a comprehensive review of Image Fusion (IF) methods, essential for combining images from diverse sources in fields like robotics, medical diagnostics, and satellite imaging. IF approaches are categorized into feature-level, pixel-level, and decision-level, each with distinct advantages and limitations. The study distinguishes between spatial domain techniques (e.g., Simple Average, PCA) and frequency domain techniques (e.g., Discrete Wavelet Transform), highlighting multi-scale methods like Laplacian Pyramid and DWT for high-quality image fusion, and addressing challenges such as misregistration and processing demands.

3 Model Architecture

This section details the design of the proposed model, covering feature extraction, attention mechanisms for plant disease categorization, efficient edge deployment, and adaptability across crop types and imaging modalities.

3.1 Plant-Attention Transformer Model:

The proposed approach takes plant leaf images as input and applies various data augmentation techniques, such as random resizing, flipping, rotation, and color jitter, to increase the diversity of the training data. Our proposed innovative deep learning architecture (Plant-AT) combines the strengths of local and global attention mechanisms, multi-scale feature fusion, and efficient parameter utilization by controlling flow of input and output channels through the model architecture to achieve robust plant disease classification considering 8,808,068 trainable parameters. The model's architecture enables efficient real-time processing on resource-constrained edge devices. Plant-AT Model is a 3 layered architecture, essential components are shown in Fig.1:

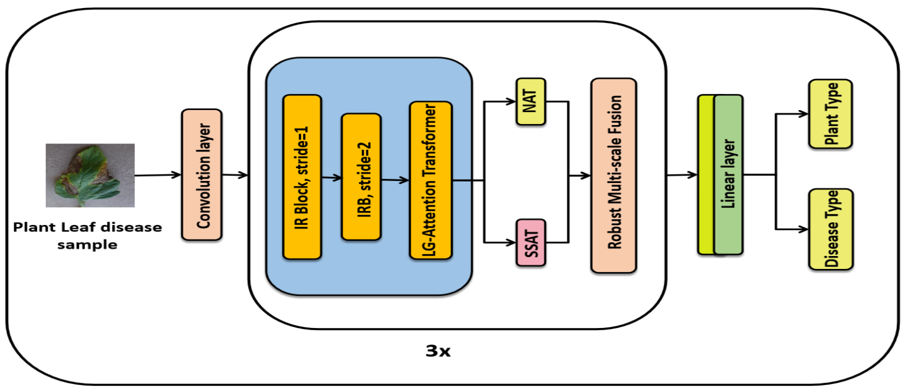


Fig. 1: Plant-AT Model: The LG-Attention Transformer (LG-AT) comprises successive branches; the Separable Self-Attention Transformer (SSAT) and the Neighborhood Attention Transformer (NAT) both iterate twice inside the LG-AT. After each layer is completed, the output is transmitted to SSAT and NAT individually, and the outputs of NAT (2 iterations) and SSAT (2 iterations) are fused using a multi-scale fusion technique.

1. **Inverted Residual Block(IRB) :** This block's core technology is depthwise separable convolution, which decreases processing cost. It serves as the basic building block for feature extraction. Let X be the input tensor with shape $H \times W \times C_{in}$.

Pointwise Convolution (1×1 Conv):

$$X_{pw1} = \text{ReLU6}(\text{Conv1} \times 1_{\text{expand}}(X)) \tag{1}$$

where $\text{Conv1} \times 1_{\text{expand}}$ is the 1×1 convolution that expands the number of channels from C_{in} to C_{exp}.

Depthwise Convolution (3×3 Depthwise Conv):

$$X_{dw} = \text{ReLU6}(\text{Conv3} \times 3\ _{\text{depthwise}}(X_{pw1})) \tag{2}$$

where $\text{Conv3} \times 3_{\text{depthwise}}$ is the depthwise convolution that applies a 3x3 filter to each channel independently.

Pointwise Convolution (1×1 Conv):

$$X_{pw2} = \text{Conv1} \times 1_{\text{project}}(X_{dw}) \tag{3}$$

where $\text{Conv1} \times 1_{\text{project}}$ is the 1×1 convolution that projects the expanded channels back to C_{out}.

Skip Connection (if applicable):

$$X_{out} = \begin{cases} X_{pw2} + X & \text{if } C_{in} = C_{out} \text{ and stride} = 1 \\ X_{pw2} & \text{otherwise} \end{cases} \tag{4}$$

$$\begin{aligned} X_{out} = {} & \text{Conv1} \times 1_{\text{project}}(\text{ReLU6}(\text{Conv3} \times 3_{\text{depthwise}} \\ & (\text{ReLU6}(\text{Conv1} \times 1_{\text{expand}}(X))))) + X \end{aligned} \tag{5}$$

An IRB with stride $= 1$ is used for image feature extraction, followed by another IRB with stride $= 2$ for spatial downsampling of feature maps, which is then sent to the LG-Attention Transformer.

2. **Local-Global Attention Transformer:** A crucial component combining global and local interdependencies within input images, which plays a vital role in increasing the recognition capacity over state-of-the-art methods, consists of two sequential branches.

Feature Unfolding: To capture global representations with spatial inductive bias, the input tensor $X_L \in \mathbb{R}^{H \times W \times d}$ is unfolded into n non-overlapping flattened patches $x \in \mathbb{R}^{P \times n \times d}$, where $P = wh$, $n = \frac{HW}{P}$, and h and w are the height and width of a patch.

$$x = \text{Unfold}(X_L) \tag{6}$$

Separable Self-Attention Transformer (SSAT): It contains Layer Normalization, Separable Self-Attention, Multi-Layer Perceptron (MLP) with residual connections, which captures global interdependencies by applying self-attention with element-wise computation while maintaining spatial inductive bias. It reduces the computational cost of self-attention from $O(n^2)$ to $O(n)$ by leveraging a latent token. The unfolded features x are processed through the Separable Self-Attention mechanism [10]. Flow of essential components are shown in Fig.4:

Latent Token Creation: The latent token L is represented by the weights $W_L \in \mathbb{R}^d$ of a linear layer. This d-dimensional vector serves as a learnable reference point for computing context scores.

Context Scores with Latent Token L: The input sequence $x \in \mathbb{R}^{n \times d}$, where n is the sequence length and d is the embedding dimension, is mapped to a vector of scalars using the latent token L. This mapping computes the inner product between each token and the latent token is represented as,

$$\text{cs}_L = \text{softmax}(xW_L) \tag{7}$$

Here, $\text{cs}_L \in \mathbb{R}^n$ are the context scores with respect to the latent token L.

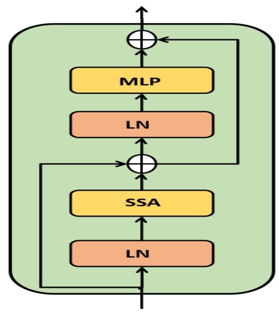


Fig. 2: SSAT

Context Vector: The context vector $\mathrm{cv}_L \in \mathbb{R}^d$ is computed as a weighted sum of the key projections $x_K = xW_K$, where $W_K \in \mathbb{R}^{d\times d}$ are the key weights:

$$\mathrm{cv}_L = \sum_{i=1}^{n} \mathrm{cs}_L(i) x_K(i) \tag{8}$$

Latent Token Usage: The latent token L enables efficient global context computation by reducing the complexity from $O(n^2)$ to $O(n)$, as each input token interacts only with L instead of all other tokens.

Final SSAT Equation: The output Y_{SSAT} of the Separable Self-Attention mechanism is obtained by combining the context scores cs_L with the key projections x_K and value projections $x_V = \mathrm{ReLU}(xW_V)$, where $W_V \in \mathbb{R}^{d\times d}$, and $W_O \in \mathbb{R}^{d\times d}$ is the output projection, given by,

$$Y_{SSAT} = \left(\sum \underbrace{\mathrm{softmax}(xW_L)}_{\mathrm{cs}_L \in \mathbb{R}^n} * xW_K \right) \cdot \mathrm{ReLU}(xW_V)W_O \tag{9}$$

Where $*$ denotes the broadcasted element-wise multiplication operation.

Feature Folding: After processing through the SSAT, the features Y_{SSAT} are folded back to obtain the final tensor $X_F \in \mathbb{R}^{H\times W\times d}$:

$$X_F = \mathrm{Fold}(Y_{SSAT}) \tag{10}$$

Dimensionality Reduction: The higher-dimensional feature maps X_F are then projected to a lower-dimensional space using a convolutional layer with a kernel size of 1×1. This operation reduces the channel dimension while preserving the spatial dimensions,

$$X_R = \mathrm{Conv}_{1\times 1}(X_F) \tag{11}$$

where $X_R \in \mathbb{R}^{H\times W\times c}$ represents the reduced-dimension feature maps, with $c < d$.

Neighborhood Attention Transformer(NAT): It Contains of Layer normalization, Neighborhood Attention, Feed forward network with residual connections. NA captures local interdependencies within the globally informed feature maps. NA is intended to allow each pixel (or token) to interact with its neighbors dynamically, effectively widening the receptive field without the need for a manually moved variation. This formulation of NA strongly resembles self-attention as the neighborhood size rises, eventually sustaining local inductive bias and translational equivariance, comparable to convolutional processes. Therefore, NA has a linear time and space complexity, while SA has a quadratic complexity. Flow of essential components are shown in Fig.3

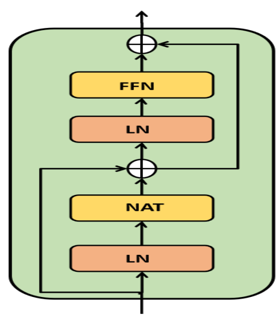


Fig. 3: NAT

Given an input $X \in \mathbb{R}^{n \times d}$, where n represents the number of tokens (e.g., pixels) and d represents the dimensionality of each token, the Neighborhood Attention is computed [8] as follows:

Linear Projections: We define the linear projections of the input as queries Q, keys K, and values V with relative positional biases $B(i, j)$ for any pair of tokens i and j.

Attention Weights: The attention weights A_i^k for the i^{th} token with neighborhood size k are computed as,

$$A_i^k = \begin{bmatrix} Q_i K_{\rho_1(i)}^T + B(i, \rho_1(i)) \\ Q_i K_{\rho_2(i)}^T + B(i, \rho_2(i)) \\ \vdots \\ Q_i K_{\rho_k(i)}^T + B(i, \rho_k(i)) \end{bmatrix} \tag{12}$$

where $\rho_j(i)$ denotes the j^{th} nearest neighbor of the i^{th} token.

Neighboring Values: The neighboring values V_i^k corresponding to the i^{th} token are defined as,

$$V_i^k = \begin{bmatrix} V_{\rho_1(i)}^T \\ V_{\rho_2(i)}^T \\ \vdots \\ V_{\rho_k(i)}^T \end{bmatrix}^T \tag{13}$$

Neighborhood Attention: The Neighborhood Attention for the i^{th} token with neighborhood size k is defined as,

$$\mathrm{NA}_k(i) = \mathrm{softmax}\left(\frac{A_i^k}{\sqrt{d}}\right) V_i^k \tag{14}$$

where $\sqrt{d}$ is a scaling factor to stabilize the gradients during training.

LG-Attention Transformer:

$$LG - AT = NAT(SSAT) \tag{15}$$

The sequential approach enhances the model's discriminative power by combining global and local feature maps. Global features capture high-level abstractions improving generalization, while local features handle fine-grained details, making the model robust to noise and small variations. This mixture of features boosts pattern recognition and generalization from training to test data. For real-time applications on resource-constrained edge devices, minimal processing overhead and rapid learning are crucial. Global features provide overall context, while local features identify specific symptoms and small-scale variations crucial for plant disease detection. Separable self-attention increases efficiency by reducing computational complexity and memory use, making it suitable for real-time applications with limited resources. The Neighborhood Attention Transformer (NAT) supports linear time complexity and 16-bit precision to maintain speed and reduce memory usage. By processing outputs from LG-Attention Transformer in parallel with NAT and Separable Self-Attention Transformer (SSAT), the architecture allows efficient feature learning and stable training by maintaining gradient flow during backpropagation. This dual-pathway approach enables dynamic feature emphasis enhancing flexibility in handling diverse plant disease data.

3. **Robust Multi-Scale Fusion Mechanism:** The suggested fusion mechanism combines the outputs of the SSAT and NAT methods to collect multi-scale information. It is used in the Plant-AT model at several different levels.In order to dynamically weigh and combine the feature maps produced by the local and global attention branches and extract the most pertinent data for the classification task, it makes use of convolutions with different kernel sizes and convolutional-attention mechanisms. Let the given input tensors X_1 (NAT) and X_2 (SSAT),

Multi-scale Convolutions:

$$Y_1 = \text{Conv}1 \times 1(X_1) + \text{Conv}1 \times 1(X_2) \tag{16}$$
$$Y_3 = \text{Conv}3 \times 3(X_1) + \text{Conv}3 \times 3(X_2) \tag{17}$$
$$Y_5 = \text{Conv}5 \times 5(X_1) + \text{Conv}5 \times 5(X_2) \tag{18}$$
$$Y_7 = \text{Conv}7 \times 7(X_1) + \text{Conv}7 \times 7(X_2) \tag{19}$$

Feature Fusion:

$$Y_{fused} = \text{ReLU}(\text{BatchNorm}(\text{Conv } 1 \times 1(\text{Concat}[Y_1, Y_3, Y_5, Y_7]))) \tag{20}$$

Channel Attention:

$$Y_{ch\_att} = Y_{fused} \cdot \sigma(\text{MLP}(\text{GAP}(Y_{fused}))) \tag{21}$$

where, GAP is Global Average Pooling, and σ is the sigmoid function.

Spatial Attention:

$$Y_{sp\_att} = Y_{ch\_att} \cdot \sigma(\text{Conv } 7 \times 7([\text{MaxPool}(Y_{ch\_att}), \text{AvgPool}(Y_{ch\_att})])) \tag{22}$$

Robust Multi-Scale Fusion:

$$Y_{final} = \text{ReLU}(Y_{sp\_att} + Y_{fused}) \tag{23}$$

The RobustMultiScaleFusion class uses convolutions with various kernel sizes (1×1, 3×3, 5×5, 7×7) to collect multi-scale information and merge features from two inputs. It uses channel and spatial attention to highlight key feature channels and significant locations in feature maps. This technique is utilized to capture both small-scale symptoms (e.g., microscopic leaf spots or early-stage fungal growth) and larger-scale patterns (e.g., overall leaf discoloration or wilting patterns). By combining local features with global context, it improves the model's capacity to deal with varied plant diseases and leaf morphologies, potentially enhancing classification accuracy.

4. **Classifier:** The final classifier consists of fully connected layers that take final output of the 3-layered Plant-AT architecture as input and passes into two separate heads for identifying plant type and disease type by the production of class probabilities using a softmax activation function.

3.2 Plant-AT Model on Edge Deployment Considerations

Despite the current lack of ONNX support for the Neighborhood Attention Transformer (NAT), alternative approaches leveraging native hardware optimizations enable the model to achieve real-time inference on Jetson AGX Orin. By efficiently utilizing the device's GPU capabilities, the model ensures low memory consumption and delivers fast inference times without relying on ONNX support. On the other hand, on highly resource-constrained devices such as Raspberry Pi, the lack of ONNX support presents a significant challenge, as it hinders the conversion of NAT into TensorFlow Lite. This limitation makes direct conversion to TensorFlow Lite infeasible, thereby restricting the deployment of the full model on CPU-only devices without substantial modifications. Future work will focus on developing an ONNX framework that supports NAT, facilitating the conversion of the model into TensorFlow Lite and enabling its deployment on low-resource devices while maintaining performance.

3.3 Plant-AT Model Adaptability

The Plant-AT is inherently designed to classify diseases across a wide range of crops, demonstrating flexibility and scalability in agricultural applications. Its adaptable architecture allows it to generalize well across different plant species and disease types, providing robust performance without the need for crop-specific retraining. Furthermore, the model's attention mechanisms, particularly the Neighborhood Attention Transformer (NAT), natively support three-dimensional (3D) data, enabling the model to process volumetric information efficiently. By modifying the Separable Self-Attention component to extend to a third dimension, as well as adapting the Robust Multi-Scale Fusion component for 3D processing, the model can seamlessly handle higher-dimensional datasets.

4 Training process and Comparison framework

This section outlines the training methodologies and performance metrics used to evaluate the model.

4.1 Performance Efficiency Metrics:

A Confusion Matrix highlights a classification model's performance by showing the number of true positive, true negative, false positive, and false negative predictions. Accuracy is the fraction of properly categorized cases in a classification model. Precision refers to the proportion of true positive forecasts among all positive predictions produced by the model, whereas recall is the fraction of genuine positive predictions among all positive instances. The F1-score is the harmonic mean of accuracy and recall. The Receiver Operating Characteristic (ROC) Curve is a graphical representation of a classifier's performance across various discrimination thresholds. It plots the True Positive Rate (TPR) against the False Positive Rate (FPR).The Area Under the Curve (AUC) summarizes the entire ROC curve into a single value, representing the classifier's overall performance. For multi-class classification, two types of averaging are used: Micro-average ROC aggregates the contributions of all classes to compute the average metric, while Macro-average ROC computes the metric independently for each class and then takes the average. The choice between micro and macro-average depends on the specific problem. Micro-average is preferred when there is class imbalance and equal instance weighting is desired. Macro-average is useful when equal importance to each class, regardless of its frequency, is required. In the context of plant disease classification, comparing both micro and macro-average

ROC-AUC provides insights into overall performance and per-class effectiveness, respectively. The Gini Coefficient is derived from the ROC curve and measures a model's ability to distinguish between classes. The Matthews Correlation Coefficient (MCC) is a valuable metric for evaluating a multiclass classifier, especially with class imbalance.

4.2 Training Process

The training and evaluation process leverages advanced techniques to optimize model performance and stability.

1. **Data Handling:** A custom dataset class efficiently loads and preprocesses plant leaf images using standardization and augmentation techniques. The WeightedRandomSampler in PyTorch addresses dataset imbalance by assigning higher weights to underrepresented classes, thereby enhancing their selection probability during training. We computed class weights inversely proportional to their frequencies and used them in the WeightedRandomSampler to ensure balanced class representation in each batch, stabilizing training and preventing model bias towards the majority class.
2. **Optimization:** We use the Adam optimizer for adaptive learning and efficiency. The objective function is a novel combined loss, integrating Cross-Entropy and Focal Loss, defined as,

$$L_{combined} = w_{CE} \cdot \left(-\sum_{i=1}^{C} y_i \log(\hat{y}_i) \right) + w_{FL} \cdot (-\alpha(1 - \hat{y}_t)^{\gamma} \log(\hat{y}_t)) \quad (24)$$

where, w_{CE} is the weight for Cross-Entropy Loss (set to 0.5), w_{FL} is the weight for Focal Loss (set to 0.5), C is the number of classes, y_i is the true probability of class i, $\hat{y}_i$ is the predicted probability of class i, α is the balancing factor for Focal Loss (set to 1), γ is the focusing parameter for Focal Loss (set to 2), $\hat{y}_t$ is the predicted probability for the true class.

   This hybrid approach addresses class imbalance by assigning greater weight to challenging examples through focal loss, while cross-entropy stabilizes learning for simpler cases. The focal loss component's adaptive response to varying class complexities enables precise gradient modulation, bolstering model resilience across underrepresented disease categories. Furthermore, fine-tuning the loss function weights offers a tailored optimization strategy, enhancing the model's performance in plant disease classification.
3. **Learning Rate Scheduling and Gradient Management:** Cosine annealing adjusts the learning rate dynamically, starting from 0.0001 and decreasing to 0.000012 by the 78^{th} epoch, to improve generalization and avoid local minima. Gradient clipping is employed to prevent exploding gradients by scaling down values beyond a set threshold, ensuring training stability in deep networks. These strategies collectively enhance the model's performance in navigating high-dimensional parameter spaces and processing data effectively.

5 Dataset

The Plant Village Dataset, containing images of two plant types(tomato and grape) was used to train the Plant-AT model. Tomato has 10 disease classes (Bectarial spot, Early blight, Healthy, Target spot, Late blight, Leaf mold, Mosaic virus, Yellow leaf curl virus, Spider mites, and Septoria leaf spot), each with 1000 images. Grape has four classes (Black Rot, Healthy, Late blight, and Esca). In validation, tomato disease classes consist of 100 images each class, whereas grape disease classes consist of 104 images in each class. While the initial experiments focused on tomato and grape datasets, the Plant-AT is designed to generalize across diverse crops and disease types.

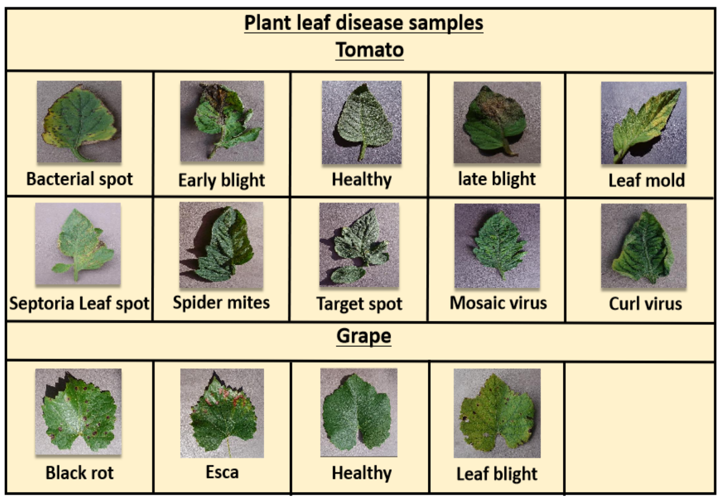


Fig. 4: Plant leaf disease samples

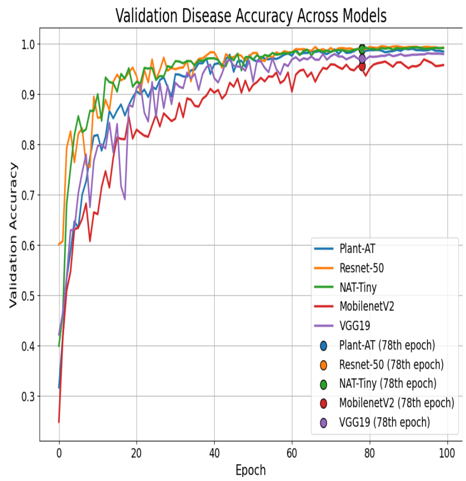
6 Results and Discussions

We evaluate the performance of the proposed framework against state-of-the-art methodologies, analyzing key performance indicators to discern the strengths and weaknesses of each approach. This assessment is conducted using a meticulously curated set of hyperparameters, as detailed in Table 1, which have been optimized to improve convergence, model accuracy, and overall robustness.

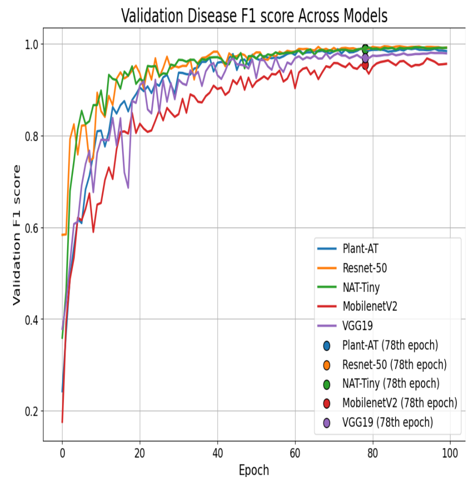
Table 1: Hyperparameters used in the training loop

Hyperparameter	**Value**
Image size	224×224
Batch Size	64
Number of Epochs	100
Patience for Early Stopping	15
Number of Workers	4
Learning Rate (lr)	0.0001
Weight Decay	1e-4
Scheduler	CosineAnnealingLR (T_max=100)
Gradient Clipping	1.0
Loss Weighting	Plant loss + 3 * Disease loss

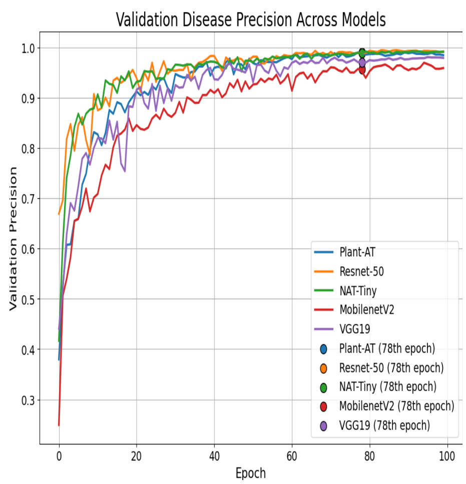
Fig. 5: Validation Graph of Disease Classification Performance Across Models



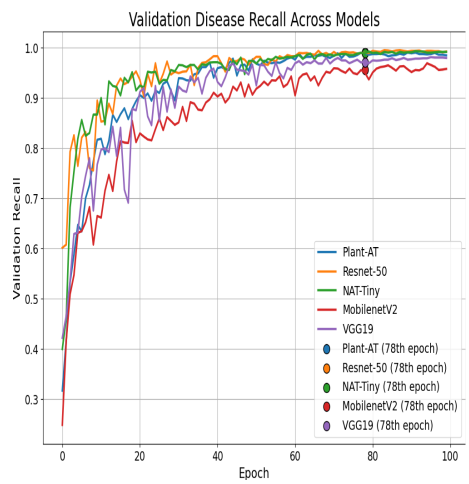
(a) Validation disease accuracy



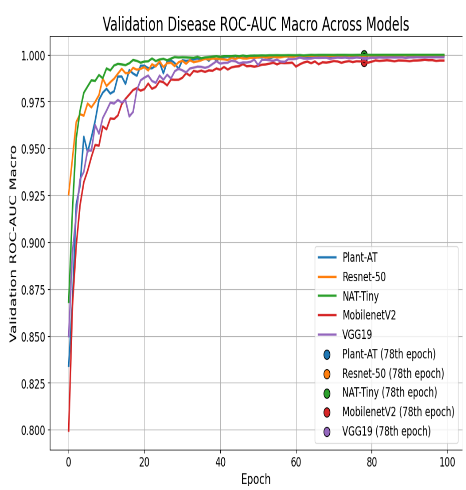
(b) Validation disease F1 score



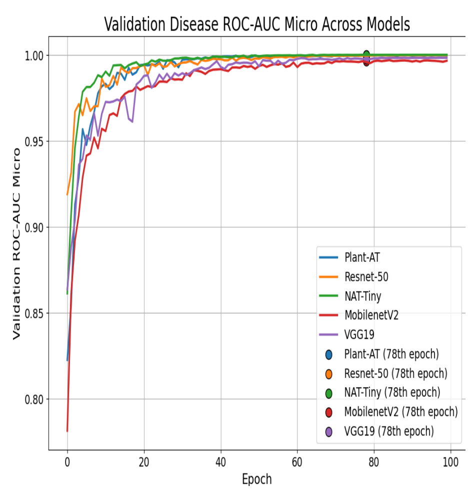
(c) Validation disease precision



(d) Validation disease Recall

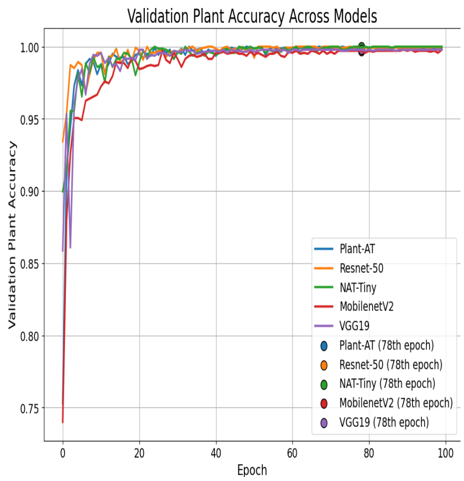


(e) Validation disease ROC-AUC Macro

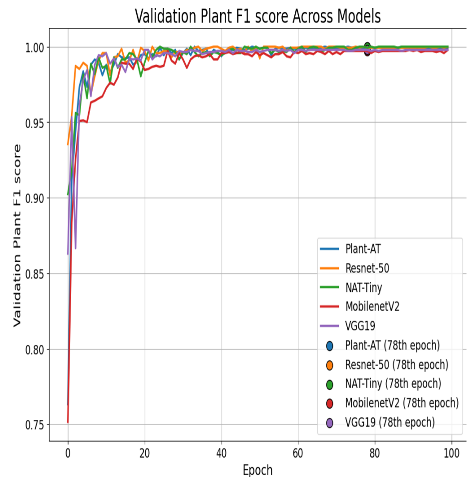


(f) Validation disease ROC-AUC Micro

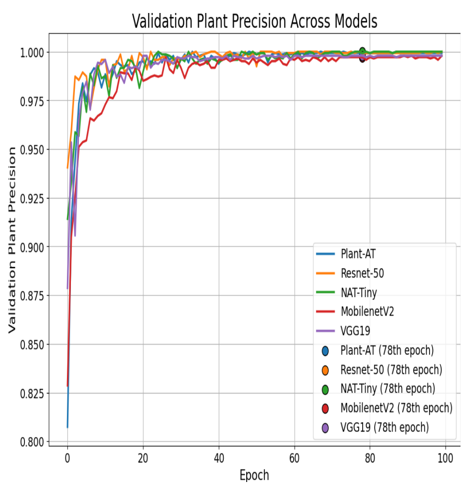
Fig. 6: Validation(Val) Graph of Plant Classification Performance Across Models



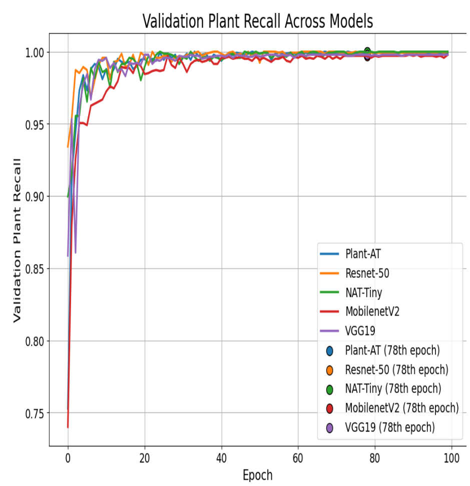
(a) Validation plant type accuracy



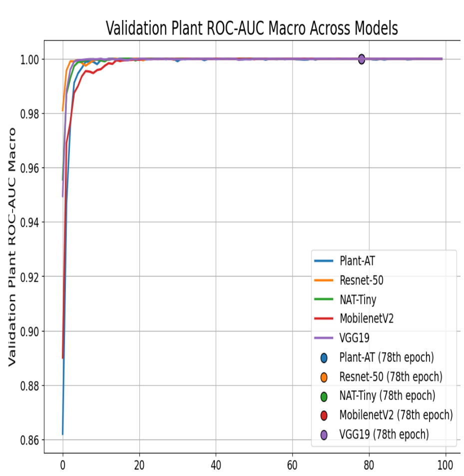
(b) Validation plant type F1 score



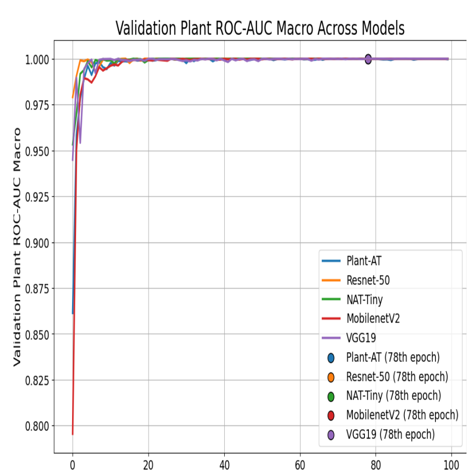
(c) Validation plant type precision



(d) Validation plant type Recall



(e) Val plant type ROC-AUC Macro



(f) Validation plant type ROC-AUC Micro

Table 2: Model Performance on the PlantVillage Dataset

Section 1: Model Performance on PlantVillage Dataset (Disease Type)								
Model	**Parameters**	**Accuracy**	**Precision**	**Recall**	**F1-score**	**AUC**	**MCC**	**Gini**
NAT-Tiny	42,447,184	98.94%	98.97%	98.94%	98.94%	0.9999	0.9886	0.9998
ResNet-50	23,540,816	98.94%	98.96%	98.94%	98.94%	0.9994	0.9886	0.9988
VGG-19	139,635,792	97.03%	97.12%	97.03%	97.01%	0.9975	0.9682	0.9950
MobileNetV2	3,884,048	95.55%	95.77%	95.55%	95.48%	0.9961	0.9524	0.9922
Plant-AT	8,808,068	99.01%	99.06%	99.01%	99.01%	0.9998	0.9894	0.9997
Section 2: Model Performance on PlantVillage Dataset (Plant Type)								
Model	**Parameters**	**Accuracy**	**Precision**	**Recall**	**F1-score**	**AUC**	**MCC**	**Gini**
NAT-Tiny	42,447,184	99.97%	99.97%	99.97%	99.97%	1.0000	1.0000	1.0000
ResNet-50	23,540,816	99.93%	99.93%	99.93%	99.93%	1.0000	1.0000	1.0000
VGG-19	139,635,792	99.79%	99.79%	99.79%	99.79%	0.9999	0.9949	0.9998
MobileNetV2	3,884,048	99.72%	99.72%	99.72%	99.72%	0.9932	0.9999	0.9999
Plant-AT	8,808,068	99.93%	99.93%	99.93%	99.93%	1.0000	0.9983	1.0000

7 Conclusion

The Plant-AT model demonstrates superior performance across multiple performance metrics compared to previous state-of-the-art techniques. Its adaptable design is well-suited for diverse agricultural applications and optimized for deployment on resource-constrained devices, meeting essential requirements for edge implementation. The integration of advanced imaging techniques significantly enhances the model's capacity to address critical agricultural tasks. This approach equips farmers with data-driven insights for irrigation, fertilization, and pest management, ultimately leading to higher crop yields and a reduced environmental footprint. Additionally, the model effectively addresses challenges in data integration and real-time processing, ensuring its practicality for in-field applications. Overall, this model contributes to a more efficient, sustainable, and productive agricultural sector, playing an essential role in addressing global food demands while minimizing environmental impact.

8 Acknowledgement

We gratefully acknowledge the support provided by the Science and Engineering Research Board (SERB) of India through funding under the e-Krushi Precision Agriculture initiative. This assistance has been instrumental in advancing our research on precision agriculture technologies.

References

1. A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Neural Inf. Process. Syst. (NIPS)*, 2017.

2. A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.
3. W. Shafik, A. Tufail, A. Namoun, L. C. D. Silva, and R. A. Awg H. Mohd Apong, "A systematic literature review on plant disease detection: Motivations, classification techniques, datasets, challenges, and future trends," *IEEE Access*, vol. 11, 2023.
4. S. Coulibaly, B. Kamsu-Foguem, D. Kamissoko, and D. Traore, "Deep learning for precision agriculture: A bibliometric analysis," *Intelligent Systems with Applications*, vol. 16, pp. 200102, 2022.
5. A. Khan, Z. Rauf, A. Sohail, A. R. Khan, H. Asif, A. Asif, and U. Farooq, "A survey of the vision transformers and their CNN-transformer based variants," *Artificial Intelligence Review*, vol. 56, no. Suppl 3, pp. 2917-2970, 2023.
6. L. Papa, P. Russo, I. Amerini, and L. Zhou, "A survey on efficient vision transformers: algorithms, techniques, and performance benchmarking," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.
7. A. Hassani, S. Walton, J. Li, S. Li, and H. Shi, "Neighborhood attention transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 6185-6194.
8. A. Hassani, "Neighborhood Attention: Dynamic Restriction of Self Attention," Department of Computer Science, University of Oregon, Eugene, OR, USA, Tech. Rep., 2023.
9. Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 10012-10022.
10. S. Mehta and M. Rastegari, "Separable self-attention for mobile vision transformers," *arXiv preprint arXiv:2206.02680*, 2022.
11. H. Kaur, D. Koundal, and V. Kadyan, "Image fusion techniques: a survey," *Archives of Computational Methods in Engineering*, vol. 28, no. 7, pp. 4425-4447, 2021.
12. Poornima Singh Thakur, Shubhangi Chaturvedi, Pritee Khanna, Tanuja Sheorey, and Aparajita Ojha. "Vision transformer meets convolutional neural network for plant disease classification." *Ecological Informatics* 77 (2023): 102245.
13. Wasi Ullah, Kashif Javed, Muhammad Attique Khan, Faisal Yousef Alghayadh, Mohammed Wasim Bhatt, Imad Saud Al Naimi, and Isaac Ofori. "Efficient identification and classification of apple leaf diseases using lightweight vision transformer (ViT)." *Discover Sustainability* 5, no. 1 (2024): 116.